

Develop a predictive text system using N-Gram and HMM (Hidden Markov Models) to predict the next word based on previous context.

Aim:

To develop a predictive text system using N-Gram and HMM (Hidden Markov Models) to predict the next word based on previous context.

Dataset: BrownCorpus

Step 1: Load the dataset

```
import nltk

nltk.download('brown')

from nltk.corpus import brown

sentences = brown.sents()
```

Step 2: Convert text to lowercase (remove special characters, punctuation, extra spaces)

```
import re

clean_sentences = []

for sent in sentences:

    text = " ".join(sent).lower()

    text = re.sub(r'^a-z\s', "", text)

    text = re.sub(r'\s+', ' ', text).strip()

    clean_sentences.append(text)
```

Step 3: Tokenization (sentence & word)

```
tokenized_sentences = []

for sent in clean_sentences:

    tokenized_sentences.append(sent.split())
```

Step 4: Create Vocabulary

```
vocab = set()

for sent in tokenized_sentences:

    for word in sent:

        vocab.add(word)

vocab_size = len(vocab)
```

N-GRAM MODEL

Step 5: Define N-gram (Bigram & Trigram):

```
from nltk.util import ngrams

bigrams = []

trigrams = []

for sent in tokenized_sentences:

    if len(sent) >= 2:

        bigrams.extend(list(ngrams(sent, 2)))

    if len(sent) >= 3:

        trigrams.extend(list(ngrams(sent, 3)))
```

Step 6: Generate sentence using N-gram:

```
generated_sentence = ['the']

for i in range(10):

    last_word = generated_sentence[-1]

    candidates = []

    for bg in bigrams:

        if bg[0] == last_word:

            candidates.append(bg[1])

    if len(candidates) == 0:

        break

    generated_sentence.append(candidates[0])

generated_sentence
```

o/p:

```
['the',
 'fulton',
 'county',
 'grand',
 'jury',
 'said',
 'friday',
 'an',
 'investigation',
```

```
'of',  
'atlantas']
```

Step 7: Count word frequency:

```
from collections import Counter  
  
unigram_freq = Counter()  
  
bigram_freq = Counter()  
  
for sent in tokenized_sentences:  
  
    unigram_freq.update(sent)  
  
    bigram_freq.update(list(ngrams(sent, 2)))
```

Step 8: Compute N-gram probability (MLE)

```
word1 = 'the'  
  
word2 = 'man'  
  
mle_prob = bigram_freq[(word1, word2)] / unigram_freq[word1]  
  
mle_prob
```

o/p:

```
0.00265824412971088
```

Step 9: Apply Laplace smoothing

```
laplace_prob = (bigram_freq[(word1, word2)] + 1) / (unigram_freq[word1] + vocab_size)  
  
laplace_prob
```

o/p:

```
0.0016086575021936238
```

Step 10: Predict next word

```
input_word = 'the'  
  
max_prob = 0  
  
next_word = None  
  
for v in vocab:  
  
    prob = (bigram_freq[(input_word, v)] + 1) / (unigram_freq[input_word] + vocab_size)  
  
    if prob > max_prob:
```

```
max_prob = prob
```

```
next_word = v
```

```
next_word
```

o/p:

```
'first'
```

Step 11: Generate output

```
ngram_output = ['the', next_word]
```

```
ngram_output
```

o/p:

```
['the', 'first']
```

Accuracy for N-gram next-word prediction:

```
correct = 0
```

```
total = 0
```

```
for sent in tokenized_sentences[:1000]:
```

```
    if len(sent) < 2:
```

```
        continue
```

```
    prev_word = sent[-2]
```

```
    actual_word = sent[-1]
```

```
    max_prob = 0
```

```
    predicted_word = None
```

```
    for v in vocab:
```

```
        prob = (bigram_freq[(prev_word, v)] + 1) / (unigram_freq[prev_word] + vocab_size)
```

```
        if prob > max_prob:
```

```
            max_prob = prob
```

```
            predicted_word = v
```

```
    if predicted_word == actual_word:
```

```
        correct += 1
```

```
    total += 1
```

```
ngram_accuracy = correct / total
```

```
ngram_accuracy
```

o/p:

0.10040567951318459

HIDDEN MARKOV MODEL:

Step 5: Define HMM components

```
observations = vocab
```

```
transition = {}
```

```
emission = {}
```

```
states = set()
```

Step 6: POS Tagging (Hidden States)

```
nltk.download('averaged_perceptron_tagger')
```

```
tagged_sentences = []
```

```
for sent in tokenized_sentences[:5000]:
```

```
    tagged_sentences.append(nltk.pos_tag(sent))
```

Step 7: Estimate Transition & Emission Counts

```
from collections import Counter
```

```
transition_count = Counter()
```

```
emission_count = Counter()
```

```
state_count = Counter()
```

```
for sent in tagged_sentences:
```

```
    prev_tag = None
```

```
    for word, tag in sent:
```

```
        states.add(tag)
```

```
        state_count[tag] += 1
```

```
        emission_count[(tag, word)] += 1
```

```
        if prev_tag is not None:
```

```
            transition_count[(prev_tag, tag)] += 1
```

```
        prev_tag = tag
```

Step 8: Calculate Transition & Emission Probabilities

```
for (t1, t2) in transition_count:
```

```
    transition[(t1, t2)] = transition_count[(t1, t2)] / state_count[t1]
```

```
for (tag, word) in emission_count:
```

```
    emission[(tag, word)] = emission_count[(tag, word)] / state_count[tag]
```

Step 9: Train HMM Model

```
model_states = list(states)
```

Step 10: Predict Next Word

```
input_sentence = ['the', 'man']
```

```
last_word = input_sentence[-1]
```

```
candidate_words = {}
```

```
for (tag, word) in emission:
```

```
    if word == last_word:
```

```
        for next_tag in model_states:
```

```
            if (tag, next_tag) in transition:
```

```
                for w in observations:
```

```
                    if (next_tag, w) in emission:
```

```
                        candidate_words[w] = transition[(tag, next_tag)] * emission[(next_tag, w)]
```

```
hmm_next_word = max(candidate_words, key=candidate_words.get)
```

```
hmm_next_word
```

Step 11: Generate Output

```
hmm_output = input_sentence + [hmm_next_word]
```

```
hmm_output
```

Accuracy for HMM next-word prediction (Emission-based)

```
correct = 0
```

```
total = 0
```

```
for sent in tagged_sentences[:500]:
```

```
    if len(sent) < 2:
```

```
        continue
```

```
    prev_word, prev_tag = sent[-2]
```

```
    actual_word, actual_tag = sent[-1]
```

```
    predictions = {}
```

```
    for w in vocab:
```

```
if (prev_tag, w) in emission:

    predictions[w] = emission[(prev_tag, w)]

if len(predictions) == 0:

    continue

predicted_word = max(predictions, key=predictions.get)

if predicted_word == actual_word:

    correct += 1

total += 1

hmm_accuracy = correct / total

hmm_accuracy
```

O/P:

0.0

Result:

N-gram model achieved about **10% accuracy**, as it predicts the next word based on frequent word sequences in the corpus. HMM model showed **0% accuracy**, since it predicts words based on POS patterns, which rarely match the exact next word in the dataset.